

L AFC-Evict: A Large-Scale Counterfactual Benchmark for Learned Cache Eviction

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Abstract

Learned cache-eviction research has outgrown small ad hoc evaluations, yet empirical comparison remains difficult because prior studies often rely on private traces, release-incomplete preprocessing pipelines, or supervision targets that are not comparable across papers. This paper presents L AFC-Evict, a benchmark package for learned cache eviction in caching systems built around decision-aligned counterfactual supervision. Each canonical candidate row represents one candidate victim considered at one full-cache miss decision and includes release-facing provenance fields, cache-state features, and a finite-horizon counterfactual label. The canonical loss label, `y_loss`, is the continuation-policy miss cost over a finite horizon after forcing that candidate eviction. The corresponding value label, `y_value`, is defined as $-\text{y_loss}$. The benchmark also exposes decision-level and pairwise benchmark views to support value prediction, best-candidate selection, pairwise preference learning, and label-surface characterization.

The evaluated open-release snapshot covers five selected trace families—cloudphysics, metacdn, metakv, twemcache, and wiki2018—and its release metadata record 277,995,072 candidate rows, 2,363,286 decision-view rows, and a capped pairwise sample of 1,000,000 rows. Two additional families, brightkite and citibike, remain excluded from the open subset. Beyond benchmark definition, release schema, validation, and reproducibility, the paper reports full candidate-label summaries, a linear `y_loss` regression baseline, a best-candidate selector induced by that linear score, and pairwise-sample-backed feature baselines. Across all candidate rows, `y_loss` has

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mean 7.7561, standard deviation 4.6542, and 5/25/50/75/95th percentiles of 2/4/7/11/16; the mirrored `y_value` summary confirms the sign-flipped construction. The linear regression baseline is intentionally weak (test MAE 3.8739, RMSE 4.7152, $R^2 = 0.0026$), while the derived best-candidate selector reaches 0.8753 optimal-selection rate with 0.1251 mean regret. The shipped pairwise sample’s A/B orientation is assigned by a deterministic hash over stable decision and pair-identity fields rather than by candidate-identifier ordering, giving a balanced non-tie split (about 49.8%/50.2%); on the resulting non-tie subset (121,738 of 1,000,000 rows), a logistic feature-difference baseline reaches 0.9660 accuracy with 0.0824 log loss after joining released candidate features back to the capped sample for evaluation only. These results demonstrate benchmark executability rather than online policy quality, and they are not evidence of online-policy superiority. The labels are therefore interpreted as decision-aligned finite-horizon counterfactual targets rather than offline-optimal supervision.

Keywords: learned cache eviction, benchmark datasets, counterfactual supervision, cache replacement

1. Introduction

When a cache request misses and the cache is full, a system must pick one resident object to evict before admitting the requested object. Evaluating whether a given eviction decision was a good one requires knowing what would have happened to each candidate victim if it, rather than another object, had been evicted – information that is not directly observable from a single trace replay, because only one candidate is actually evicted at each decision. This paper studies a reusable way to construct that counterfactual evidence at scale, and evaluates when it is informative, when it degenerates, how it relates to actual closed-loop cache performance, and how sensitive it is to one of its own modeling assumptions.

L AFC-Evict materializes, for real cache traces, a per-decision, per-candidate finite-horizon counterfactual miss cost: for each object that could have been evicted at a full-cache miss, how many additional misses would occur over the next H requests if that object, specifically, had been the one evicted. This candidate-level quantity is stored as `y_loss` (a convenience alias `y_value = -y_loss` is also provided; see Section 7). Section 2 works through one full decision by hand – resident objects, a forced eviction, a future request

window, and the resulting candidate costs, including a case where two candidates tie for optimal – before any aggregate statistic is introduced, so that every quantity used later in the paper is grounded in a concrete example first.

The contribution of this paper is not a new eviction policy. It is an evaluation of a reusable offline supervision surface against closed-loop cache-policy behavior, organized around five questions. **(RQ1)** Is the finite-horizon counterfactual target actually discriminative between candidates, and when does it degenerate to an uninformative tie? We find that it is not uniformly discriminative: aggregate tie rates are substantial, and no audited decision has a strict unique winner, but discriminativeness varies strongly and predictably by workload, cache capacity, and horizon, and one workload in our evaluation is a genuine, mechanistically explained negative control. **(RQ2)** Does the offline candidate ranking correspond to actual closed-loop cache performance? Across a production-scale closed-loop evaluation, offline and closed-loop policy orderings agree in the large majority of workload/ capacity regimes where the offline signal itself is not near zero, with two specific, characterized exceptions. **(RQ3)** How does this correspondence depend on workload, capacity, and horizon? We find that capacity-scale locality in the underlying trace predicts both offline discriminativeness and closed-loop policy separation, and that a horizon of 16 future requests aligns offline and closed-loop evidence more strongly than shorter horizons, though modestly. **(RQ4)** How sensitive are the labels to the choice of continuation policy after a forced eviction? A sampled robustness study at two cache capacities finds the labels robust to two alternative continuation policies by a pre-registered criterion, while also showing that this robustness is not absolute at the level of individual decisions. **(RQ5)** What does this imply in practice for a researcher deciding between offline and closed-loop evaluation? We argue, and the evidence in this paper supports, that the offline supervision this dataset provides complements rather than replaces closed-loop evaluation, and we give concrete, evidenced guidance for when the offline signal can be trusted.

Every quantitative claim in this paper traces to a specific, independently validated evidence artifact; Sections 7 through 8 and the limitations discussion in Section 9 state the exact scope, sample sizes, and caveats attached to each result rather than presenting them as unqualified findings. The underlying release also exposes candidate-level, decision-level, and pairwise derived views of the same supervision at production scale (Table 1), which supports the evaluation methodology in this paper and is also intended for

reuse by other researchers studying learned cache eviction; release construction, schema, and validation are described in Sections 3–6 but are not the paper’s central argument.

To our knowledge, existing public cache-replacement resources do not provide a reusable candidate-level counterfactual supervision surface at this scale together with a direct evaluation of its relationship to closed-loop cache-policy behavior; we make this claim with the restrained scope stated above, not as a claim of exhaustive prior-art coverage (see Section 10).

The rest of the paper proceeds as follows. Section 2 gives the worked example and motivates decision-aligned counterfactual supervision. Sections 3–6 describe construction, label definition, schema, and release validation. Section 7 defines the benchmark tasks. Section 8 reports target discriminativeness (RQ1) and, together with material drawn from the closed-loop production evaluation, addresses RQ2–RQ4 (offline/ closed-loop correspondence, workload/capacity/horizon dependence, and continuation-policy robustness); practical implications (RQ5) are synthesized alongside it. Section 9 states limitations, ethics, and reuse constraints, followed by related work and conclusion.

2. Background and Motivation: Learned Cache Eviction

At each full-cache miss, a caching system must evict one resident object before admitting the requested one. Classical policies such as LRU operationalize this choice through simple recency structure, while learned or learning-augmented approaches attempt to predict which eviction will lead to lower future miss cost. In practice, these approaches differ in feature sets, prediction targets, and evaluation methodology.

The main difficulty for a benchmark is that victim selection is structured, not i.i.d. A model does not act on one object in isolation; it acts within a decision set whose size, feature spread, and ambiguity vary across traces and capacities. A useful benchmark therefore preserves decision boundaries instead of flattening them away.

This observation motivates the core LAFC-Evict design choice: represent supervision at the candidate-within-decision level. Each row corresponds to one candidate victim at one full-cache eviction decision, and the label records the finite-horizon counterfactual miss cost of forcing that eviction under a fixed continuation policy. This representation allows both pointwise learning

and within-decision evaluation, while keeping the supervised object aligned with the actual cache-eviction choice.

From that candidate-level foundation, the benchmark derives two secondary evaluation surfaces. Grouping all rows with the same `decision_id` produces a decision view that records candidate count, optimal-set membership, tie structure, and regret summaries. Sampling candidate pairs from the same decision produces the shipped pairwise view, which turns within-decision comparisons into a simpler classification problem without discarding the original decision boundary.

A single full-cache miss can therefore appear in three complementary forms. If one decision exposes k admissible victims, the release records k candidate rows, one decision-group summary, and up to $k(k-1)/2$ within-decision pairwise comparisons before any sampling cap is applied. The manuscript keeps that task-view progression textual rather than introducing a second overview figure beside the main pipeline diagram.

3. Benchmark Design

L AFC-Evict is organized around five layers: external raw traces, processed traces, generated features, generated labels, and benchmark tasks. The release package begins after candidate-row generation and focuses on converting existing generated data into a reusable benchmark package for cache-eviction benchmarking, with schema, validation, and release metadata.

The evaluated open subset is `lafc-evict-v0.1-open`. The evaluated release metadata identifies five selected families: `cloudphysics`, `metacdn`, `metakv`, `twemcache`, and `wiki2018`. The family registry also lists `brightkite` and `citibike` as excluded from the evaluated open subset for redistribution reasons. This scope boundary is a release-governance decision, not a statement that the broader generated dataset mix is automatically redistributable. Not every selected family appears in every split in the evaluated release; for example, `metacdn` contributes train and validation decision rows but no test decision rows. Split assignment is produced upstream of this release package, which only normalizes and validates split values rather than assigning them, so uneven per-family split coverage reflects that upstream construction rather than a benchmark design guarantee of uniform cross-family generalization.

The canonical benchmark unit is the candidate row. A single row corresponds to one candidate victim at one full-cache miss decision, so one eviction decision with $|C_t|$ admissible victim choices contributes $|C_t|$ candidate

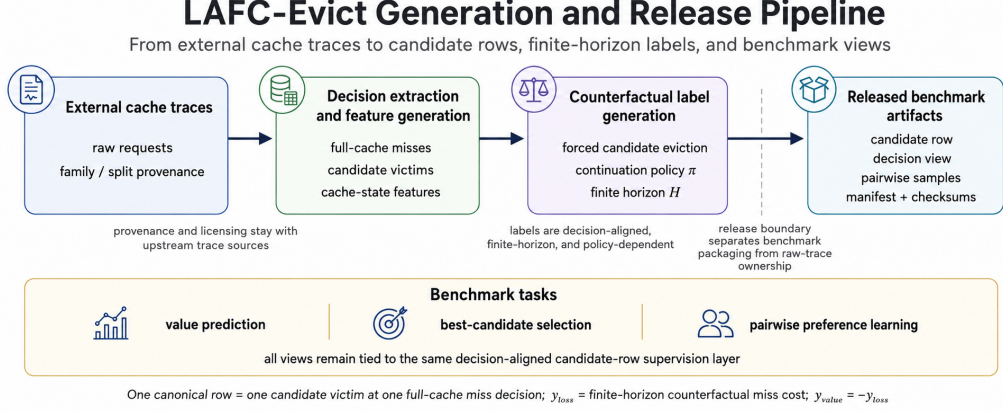


Figure 1: L AFC-Evict generation and release pipeline. The benchmark converts external cache traces into decision-level candidate rows, finite-horizon counterfactual labels, and released benchmark views for value prediction, best-candidate selection, and pairwise preference learning.

rows. Decision-level and pairwise benchmark objects are derived from that candidate-level base rather than collected independently. This design keeps the benchmark centered on cache-eviction decisions instead of conflating the benchmark contribution with end-to-end deployment claims.

The benchmark is designed to satisfy four requirements:

1. **Decision alignment.** Preserve within-decision structure and candidate competition.
2. **Task multiplicity.** Support pointwise, decision-level, and pairwise learning tasks from one release.
3. **Releaseability.** Separate public benchmark artifacts from upstream raw-trace ownership and hosting constraints.
4. **Validation.** Expose checks for schema, consistency, family scope, and checksums.

4. Counterfactual Label Generation

The benchmark label is defined on candidate victims rather than on completed policies. Let $r_1, r_2, \dots$ denote a request sequence, and let t index a

Table 1: Dataset scale from preserved-release metadata. Counts are manifest-backed or view-backed as noted.

metric	value
candidate rows	277,995,072
decision rows	2,363,286
pairwise-sample rows	1,000,000
preserved release size	2.7G
candidate parquet files	168
total release files	176
manifest file_inventory count	176
checksum entry count	175
selected trace families	cloudphysics, metacdn, metakv, twemcache, wiki2018
excluded trace families	brightkite, citibike
full validation status	passed

Table 2: Split counts from manifest-backed candidate-row metadata. These are candidate-row counts, not decision-view counts.

split	row_count
test	28,205,952
train	205,726,368
val	44,062,752

request r_t that misses when the cache is full. Write S_t^- for the cache state immediately before the eviction choice; the candidate set $C_t = S_t^-$ is every resident object, so $|C_t|$ equals the cache capacity (32, 64, 128, or 256 objects in the evaluated release), counted in objects rather than bytes, with upstream object sizes not separately modeled. For any candidate $c \in C_t$, consider the intervention that evicts c , admits the missed request into the freed slot, and then follows a fixed continuation policy π for the next H requests, resolving every subsequent miss according to π ; in the evaluated release, π is LRU continuation rather than a Belady/offline-optimal rule. The resulting finite-horizon counterfactual loss counts those subsequent misses, so every miss contributes unit cost:

$$L_{t,H}^\pi(c) = \#\{\tau \in \{t+1, \dots, t+H\} : r_\tau \text{ misses in the induced trajectory}\}.$$

The release stores this quantity as `y_loss`, and it stores the corresponding value label as `y_value = -y_loss`. The triggering miss at time t is common to all candidates and is therefore not the decision-discriminating part of the label.

This label has two consequences for the paper. First, it creates a common numerical target for regression-style benchmark tasks. Second, it induces structured decision-level objects. For a fixed decision (t, S_t^-, C_t) , the benchmark-optimal candidate set is

$$C_t^* = \arg \min_{c \in C_t} L_{t,H}^\pi(c),$$

and the per-candidate regret is

$$\Delta_{t,H}^\pi(c) = L_{t,H}^\pi(c) - \min_{c' \in C_t} L_{t,H}^\pi(c').$$

These quantities define best-candidate selection targets, pairwise preferences, and tie structure within a decision.

This supervision is a generated counterfactual label family, not an upstream annotation. It is finite-horizon, not asymptotic. It is tied to a continuation policy, not a universally optimal control objective. The label is therefore decision-aligned for the benchmarked eviction choice, but it is not an offline-optimal label in the sense of solving the full future control problem. These boundaries are central to the scientific positioning of the paper.

5. Dataset Schema and Views

The canonical candidate schema currently consists of decision metadata, candidate identity, split information, labels, and feature columns. Each row corresponds to exactly one candidate victim at exactly one full-cache miss decision. Trace identity and split metadata are recorded through `trace_name`, `trace_family`, `dataset_source`, and `split`. Cache configuration is recorded through `capacity` and `horizon`. Decision identifiers include `decision_id` and `decision_t`. The schema also records a per-decision chunk ID. That field is named `decision_chunk_id`. Candidate identity is represented through the field `candidate_page_id`. The labels are `y_loss`, the finite-horizon counterfactual miss cost after forcing that candidate eviction under the fixed continuation policy, and `y_value = -y_loss`. Feature columns cover request context, candidate context, predictor/LRU signals, cache summary statistics, and recent request or hit rates.

Table 3: Compact summary of candidate-row schema groups and derived-view fields.

Column group	Notes
Decision metadata	family, capacity, horizon, split, IDs
Candidate identity	candidate page ID
Labels	<code>y_loss</code> , <code>y_value</code>
Feature groups	request, candidate, cache, disagreement, rate
Derived decision fields	tie and regret summaries
Derived pairwise fields	A/B labels and regret differences

Beyond candidate rows, the release defines two derived views. The *decision view* aggregates all rows that share a decision identifier and summarizes each decision with candidate count, minimum loss, maximum value, the set of optimal candidate IDs, tie count, and regret summary statistics. The *pairwise view* compares candidate pairs from the same decision and derives pairwise preference labels from the ordering of their `y_loss` values, with ties when the losses are equal. The evaluated open release does not ship the full pairwise view; it ships a capped pairwise sample of 1,000,000 rows instead.

Operationally, the schema can be read as a layered projection over the same decision events: candidate rows are the atomic supervised objects, the decision view groups those rows by `decision_id`, and the shipped pairwise sample keeps only selected within-decision comparisons. Table 3 summarizes those column groups and derived fields directly in manuscript form.

6. Release Construction and Validation

The release pipeline is designed to package existing generated candidate rows into a benchmark artifact without requiring raw traces in the release package. The builder uses an out-of-core DuckDB path to ingest candidate shards, normalize split values, emit partitioned Parquet candidate rows, derive the decision view, optionally create a capped pairwise sample, and write release metadata and checksums.

The validation surface is part of the benchmark contribution. The validation workflow includes:

- schema completeness for candidate partitions,
- split normalization and split-set consistency,
- presence of selected families and absence of blocked families,

Table 4: Release-package layout using release-relative paths for the evaluated benchmark snapshot.

artifact	relative_path
candidate rows	data/candidate_rows/
decision view	data/decision_view/decision_view.parquet
pairwise sample	data/pairwise_sample/pairwise_sample.parquet
release manifest	metadata/release_manifest.json
checksums	metadata/checksums.sha256

- duplicate candidate detection within decision groups,
- consistency of decision metadata within a decision,
- decision-view schema and count consistency,
- checksum verification.

The evaluated open-release snapshot passed metadata repair, the provided verification suite, publication-bundle validation, and end-to-end release validation. The validation evidence comes from manifest-backed counts, schema checks, split checks, duplicate detection, and checksum verification implemented in reproducible scripts. The manuscript therefore reports validation of the release package without relying on unpublished hosting details. It does not claim that final public hosting or DOI assignment is complete. It also does not treat superseded status notes as the authoritative source of scientific evidence.

Table 4 summarizes the release-package components used in the evaluation: candidate rows, derived benchmark views, manifest metadata, and checksums. Validation claims rely on those package contents together with reproducible validation scripts rather than on any standalone status note.

7. Benchmark Tasks and Baselines

LAFC-Evict supports four benchmark tasks from one release.

Candidate-level value prediction.. Predict `y_loss` or `y_value` from a single candidate row. This is a pointwise task over the canonical benchmark unit and measures how well a model predicts finite-horizon counterfactual eviction quality for one candidate at one decision.

Best-candidate selection.. Given all candidate rows for one decision, identify a member of the minimum-loss set. This task can be framed as ranking or decision-level classification and connects most directly to the actual cache-eviction choice because it evaluates whether a method selects a benchmark-optimal victim within that decision.

Pairwise preference prediction.. Predict whether candidate A is better than candidate B within the same decision, where “better” means lower `y_loss` or equivalently higher `y_value`. This task uses the pairwise view or the shipped pairwise sample and provides a simpler learning interface for within-decision preference modeling. Because equal losses induce ties, reports using this task need to state explicitly whether evaluation includes tie rows, excludes them, or maps them to a separate class. Within the shipped pairwise sample, the A/B assignment is randomized by a deterministic hash over stable decision and pair-identity fields (`orientation_method: deterministic_hash_v1`), not by candidate-identifier ordering, so the non-tie `label_a_better`/`label_b_better` split is balanced (about 49.8%/50.2%) rather than reflecting a construction-induced directional imbalance.

Candidate-label distribution characterization.. Summarize the empirical label surface without training a predictive model. This characterization task studies distributions of `y_loss`, `y_value`, regret, tie frequency, and decision ambiguity across families, capacities, horizons, and splits. It is descriptive rather than competitive, but it is important for understanding benchmark difficulty and for interpreting downstream task results.

The empirical scope of this paper remains intentionally narrow. Rather than presenting a large model bake-off, we report lightweight baselines that can be executed directly from the committed release views and that establish reproducible reference points for later work.

The current results surface covers three such references. A linear regression baseline over candidate rows predicts `y_loss` from released features. It is intentionally weak: test MAE is 3.8739 with RMSE 4.7152 and $R^2 = 0.0026$, and the validation-split R^2 is slightly negative (-0.0506). Using that same linear score for within-decision selection nevertheless provides a reproducible decision-level floor. On the pairwise task, the earlier random and majority non-tie results remain useful sanity checks, but by themselves they do not test whether the released candidate features contain decision-relevant signal.

To probe that question without changing the released benchmark schema, we build an augmented feature view for evaluation only by joining released

candidate-side features back onto the capped shipped pairwise sample. This produces a pairwise-sample-backed evaluation surface rather than a full-pairwise-universe-backed one, and no full quadratic pairwise materialization is performed. The shipped sample is itself tie-heavy: 878,262 of its 1,000,000 rows (87.8%) are exact ties, leaving 121,738 non-tie rows evaluated below. Ties remain common and are reported here as part of the label-surface characterization rather than discarded. Table 6 reports six non-tie pairwise baselines on that 121,738-row surface: the random and majority sanity checks, 1-D LRU-score and predictor-score models, a pairwise baseline induced by the released pointwise linear `y_loss` regressor, and a logistic regression over released feature differences.

The resulting evidence is stronger than the earlier sanity checks alone. The logistic feature-difference model is the strongest pairwise baseline, reaching 0.9660 accuracy with 0.0824 log loss on the non-tie subset of the shipped sample. The linear-score and LRU-score baselines also outperform the random and majority checks, which indicates that the released candidate features retain decision-relevant signal for pairwise preference prediction. These numbers nevertheless warrant conservative interpretation. They improve the empirical benchmark evidence without changing the released candidate-row, decision-view, or shipped pairwise artifacts, and they do not support full-universe or online-policy claims.

The best-candidate baseline’s evaluator previously reported 2,366,413 decision groups against a manifest-backed decision-view count of 2,363,286. That mismatch traced to a decision-key bug: the evaluator grouped candidate rows by a 6-column key instead of the canonical 9-column decision key, which double-counted 3,127 decisions whose rows were not contiguous within a partition file. The corrected evaluator now groups by the canonical 9-column decision key and reports 2,363,286 decisions, an exact match with the decision view. A cheap decision-level floor computed directly from the decision view shows the dataset is heavily tied: a uniform-random selector’s expected optimal-selection rate is 0.9912 with expected regret 0.0088, both more favorable than the linear-score selector’s observed 0.8753 optimal-selection rate and 0.1251 mean regret purely because of tie density in the label surface, not because the linear score underperforms a stronger baseline; headline selection-rate and regret numbers should be read alongside this random floor rather than in isolation.

Table 5: Pairwise-sample-backed summary statistics for the shipped sample view.

statistic	value
pairwise rows	1,000,000
split breakdown	test=102,024; train=737,896; val=160,080
family breakdown	wiki2018=250,872; cloudphysics=245,336; metakv=190,584; twemcache=169,064; metacdn=144,144
unique decisions represented	118,635
label distribution	a_better=60,673; b_better=61,065; tie=878,262
A/B orientation method	deterministic_hash_v1 (randomized, not lexicographic)

8. Empirical Characterization

The empirical characterization emphasizes benchmark structure rather than a large model bake-off. The release metadata establish the scale and heterogeneity of the benchmark, and the released candidate-row summaries add full label-distribution reporting without changing that focus.

Across 277,995,072 candidate rows, `y_loss` has mean 7.7561, standard deviation 4.6542, and 5/25/50/75/95th percentiles of 2/4/7/11/16. The corresponding `y_value` summary mirrors those values by sign, as expected from the benchmark definition. The resulting label surface is broad but still bounded by the finite horizon, which keeps both candidate-level regression errors and decision-level regret interpretable. Table 8’s exactly equal horizon counts (787,762 rows at each of horizons 4, 8, and 16) reflect the same underlying full-cache-miss decisions relabeled at each configured horizon, whereas its capacity counts correspond to genuinely different full-cache-miss events produced by replaying each trace at each cache size; pooled statistics in this paper therefore mix horizons and capacities and should be read as aggregate release summaries rather than per-configuration results.

The same result summaries also show that the simple linear-score selector often lands in the benchmark-optimal set even though its underlying value regression is weak. That combination is useful descriptive context for later benchmark users: the release is not trivial, but a lightweight heuristic already clears a reproducible lower bar. The evaluator’s grouped decision count now matches the manifest-backed decision-view count exactly (Table 10); an earlier decision-key bug that double-counted a small number of decisions has been fixed and reconciled. Because the dataset is heavily tied at the deci-

Table 6: Feature-based pairwise baselines on the non-tie subset (121,738 of 1,000,000 rows) of the shipped pairwise sample, evaluated after the A/B orientation fix (deterministic hash-based randomization, not lexicographic ordering). An evaluation-only augmented feature view joins released candidate features back to the capped sample. Results are pairwise-sample-backed rather than full-pairwise-universe-backed, and no full quadratic pairwise materialization is performed.

baseline	feature source / model type	accuracy	log loss
random non-tie	random-label sanity check	0.5006	0.6931
majority non-tie	empirical majority-class sanity check	0.5016	0.6931
LRU-score pairwise	1-D logistic on LRU-score difference	0.9225	0.2426
predictor-score pairwise	1-D logistic on predictor-score difference	0.5016	0.6931
linear-score pairwise	logistic on pointwise linear-score difference	0.9490	0.1161
logistic feature difference	logistic regression on released feature differences	0.9660	0.0824

sion level, a uniform-random selector’s expected optimal-selection rate and regret are included alongside the linear-score selector’s observed metrics so that headline numbers are not read in isolation from tie density.

9. Limitations, Ethics, and Reuse

This evaluation surface has several important constraints, and the practical guidance elsewhere in this paper should be read together with them rather than instead of them.

First, the main label is a finite-horizon, continuation-policy-specific counterfactual miss cost, not an offline-optimal label for the full future control problem. Good performance on this label does not automatically imply on-line deployment gains, which is exactly why this paper’s closed-loop evaluation and offline/closed-loop linkage analysis evaluate closed-loop behavior directly rather than relying on the offline label alone.

Second, the canonical labels use LRU as the continuation policy after a forced eviction. This paper’s continuation-policy sensitivity study reports a sampled analysis, at capacities 32 and 128 only, showing both MRU and mean-random continuation classify as robust on the pre-registered rule (median optimal-set Jaccard 1.0 at both capacities; mean cross-continuation re-

Table 7: Benchmark task definitions and representative evaluation outputs.

Task	Inputs	Representative outputs
Value regression	candidate rows	MAE, RMSE, and optional rank correlation between predicted and observed values
Best-candidate selection	decision groups	Top-1 accuracy against the benchmark-optimal set and mean realized regret relative to the best available candidate
Pairwise preference	pairwise rows from one decision	Accuracy or ROC-AUC, with tie handling stated explicitly and non-tie-only reporting marked when used
Label characterization	candidate or decision aggregates	Histograms, quantiles, tie rates, regret summaries, and per-family/capacity/horizon breakdowns

Table 8: Decision-view-backed benchmark composition by family, capacity, horizon, and split.

dimension	counts
trace_family	wiki2018=598,560; cloudphysics=576,219; metakv=454,053; twemcache=397,494; metacdn=336,960
capacity	32=613,314; 64=599,172; 128=582,678; 256=568,122
horizon	4=787,762; 8=787,762; 16=787,762
split	test=242,418; train=1,746,882; val=373,986

gret and strict reversal fractions small but not identically zero, and visibly larger under random than under MRU continuation). This robustness result is a sampled-study finding at two capacities, not a population-scale or all-capacity guarantee; a separate full-population census is in progress and is not reported here.

Third, the target is substantially tied. Across the full audited canonical dataset, the all-tied fraction is approximately 0.68 and the unique-winner fraction is exactly 0.0 – no audited decision in this dataset has a single uniquely-best candidate victim. Discriminativeness varies strongly by family, capacity, and horizon; one family, wiki2018, is a fully degenerate negative control at every capacity and horizon tested, with zero reuse events in its scored windows, which is a mathematical certainty rather than a statistical property of that trace.

Table 9: Decision-view-backed candidate-count statistics. These values summarize the number of candidates per decision in the preserved release.

group	min	med	mean	max
overall	32	64	117.63	256
cloudphysics	32	64	118.35	256
metacdn	32	64	117.12	256
metakv	32	64	119.52	256
twemcache	32	64	111.52	256
wiki2018	32	64	119.85	256

Table 10: Candidate-label and best-candidate-selection summaries from committed result JSON files.

statistic	value
candidate rows summarized	277,995,072
y_loss mean / std	7.7561 / 4.6542
y_loss q05 / q25 / q50 / q75 / q95	2 / 4 / 7 / 11 / 16
y_value mean / std	-7.7561 / 4.6542
y_value q05 / q25 / q50 / q75 / q95	-16 / -11 / -7 / -4 / -2
manifest-backed decision rows	2,363,286
best-candidate evaluator decision count	2,363,286 (exact match)
best-candidate optimal-selection rate	0.8753
best-candidate regret mean / median / max	0.1251 / 0 / 3
random-selector expected optimal-selection rate	0.9912
random-selector expected regret	0.0088

Fourth, the evaluated dataset uses unit-object, count-based capacities (32, 64, 128, or 256 objects) with unconditional admission of the requested object. Variable object sizes, byte-weighted miss cost, latency-aware cost, and dirty-write cost are not modeled by this label family, and admission is never withheld. Claims in this paper should be read as scoped to this abstraction and not generalized to byte-addressable or admission-gated cache models without further evidence.

Fifth, closed-loop evidence itself has two family-specific caveats that must be carried alongside any result that uses it. MetaCDN’s closed-loop evidence is validation-window evidence, not a temporally held-out test window (Twemcache’s is test-window). MetaKV is the only family whose scored window begins at an empty cache (a cold start, identifiable from the identity evictions = misses - capacity); this is plausibly, but not conclusively, con-

nected to MetaKV being the one Tier-1 cell where a non-LRU policy wins closed-loop outright.

Sixth, a known data-generation issue remains open: in the audited canonical data, `candidate_is_predictor_victim` is identical to `candidate_is_lru_victim` throughout, and related predictor/ bucket/confidence fields include constant or placeholder values. This paper does not make any claim that depends on those predictor fields being informative.

Seventh, benchmark reuse must preserve provenance distinctions. The release exposes generated supervision and derived views, but it does not erase the attribution, citation, or licensing requirements of the upstream raw traces from which candidate rows were derived; those raw traces remain external artifacts with their own governance constraints. The open subset is also intentionally narrower than the full generated corpus for redistribution reasons, and not every selected family appears in every split of the evaluated release package.

Finally, this paper reports closed-loop results for LRU, MRU, random, and SIEVE; it does not report a closed-loop evaluation of any learned policy trained on these labels. A frozen pilot (not the production Tier-1 evaluation reported here) previously found a simple learned policy competitive with random and SIEVE but not with LRU on one family, with the other pilot family blocked on the absence of an independent test window; neither result is repeated or extended here, and learned-policy closed-loop competitiveness remains future work rather than a claim of this paper.

10. Related Work

10.1. Trace Datasets, Workload Studies, and Benchmark Surfaces

Public traces and benchmark infrastructure provide the substrate for cache evaluation. The UMass Trace Repository and large-scale workload studies of key-value stores and Twemcache clusters demonstrate the value of realistic, heterogeneous request streams for systems research [1, 2, 3]. More broadly, benchmark artifacts such as YCSB, OLTP-Bench, and the Join Order Benchmark made comparison easier by publishing reusable workloads, interfaces, and evaluation surfaces rather than one-off experiments [4, 5, 6]. Cache-specific infrastructure such as libCacheSim and `cache_dataset` further lowers the cost of trace-driven evaluation by providing high-performance simulation support and large open trace collections [7, 8]. Existing trace libraries and simulators therefore supply essential raw material for cache studies, and

some public artifacts even expose oracle-style features or procedural oracle policies. LAFC-Evict packages a different layer of supervision: for each full-cache miss decision and candidate victim, it releases finite-horizon counterfactual loss labels under a fixed continuation policy, together with decision-level and pairwise views.

Those distinctions matter because raw traces, workload studies, and simulator-oriented artifacts do not by themselves provide a reusable table of candidate-level finite-horizon counterfactual outcomes. LAFC-Evict is intended to complement that ecosystem rather than replace it. The artifact is useful precisely when a downstream user wants to study offline value regression, best-candidate selection, preference learning, or label-distribution analysis without first regenerating the same counterfactual rollouts from scratch.

10.2. Learned Eviction and Imitation Learning

Recent learned-caching papers likewise contribute online policies, policy-selection schemes, or learning frameworks rather than reusable supervised benchmarks. LRB learns a relaxed Belady-style policy for CDN caching, LHD models hit density for key-value caches, and GL-Cache and 3L-Cache improve the efficiency and overhead profile of learned eviction at group or object granularity [9, 10, 11, 12]. LeCaR learns between replacement experts online [13], and CACHEUS extends that practical line with adaptive expert selection for cache replacement [14]. Parrot uses imitation learning to approximate Belady-style replacement decisions from past accesses [15], while Hawkeye and Mockingjay show how Belady-derived signals can guide deployable replacement policies in hardware caches [16, 17]. MAT focuses on reducing the overhead of ML-guided eviction by using a heuristic filter around a learned module [18]. ALPS emphasizes lightweight learned replacement and the practical cost of learning overheads [19], while Feedforward Neural Networks for Caching is useful precisely because it questions whether heavier learned models are always justified [20]. HR-Cache extends the learned-caching line into edge delivery settings [21]. Across these works, the primary contribution is a policy or policy-learning mechanism. By contrast, LAFC-Evict releases a static benchmark surface with candidate-level finite-horizon counterfactual labels, decision-view summaries, and a shipped pairwise sample for offline value regression, best-candidate selection, and preference-learning tasks. To our knowledge, we are not aware of a public artifact in this line that releases reusable per-candidate finite-horizon counterfactual supervision at this scale.

10.3. Modern Eviction Policies

Classical cache-replacement work defines the policy landscape that any learned benchmark must eventually interface with. Belady’s look-ahead policy remains the canonical offline reference point [22]. Offline analyses such as FOO/PFOO compute practical bounds on optimal caching with variable object sizes, but their artifacts target trace-level optimality analysis rather than a reusable per-decision, per-candidate supervised label surface [23]. Practical online policies then emphasize different signals: LIRS focuses on inter-reference recency [24], ARC adaptively balances recency and frequency [25], CLOCK-Pro revisits CLOCK-style replacement with stronger reuse sensitivity [26], and GreedyDual-Size accounts for cost and size [27]. TinyLFU and W-TinyLFU further show that admission and segmented cache management matter in modern software caches [28, 29]. More recent object-cache work shows that policy progress does not always come from heavier machinery: S3-FIFO argues that simple FIFO-based queueing can beat more elaborate baselines at scale [30], while SIEVE emphasizes turn-key deployability for web caches [31]. Those policies are strong online comparison targets for future end-to-end studies, but they are not evaluated here because LAFC-Evict is an offline benchmark paper and does not claim a new deployed eviction policy. The contribution here is reusable supervision that can be used to train or diagnose such policies, not an online replacement rule that supersedes them.

10.4. Learning-Augmented Caching and Predictive Advice

Another relevant line of work treats caching as an online problem augmented by predictions rather than as a supervised dataset. Competitive Caching with Machine Learned Advice and later learning-augmented caching results formalize how predictive advice can improve classical online algorithms while preserving robustness guarantees [32, 33]. Robust Learning-Augmented Caching: An Experimental Study extends that line empirically, while Parsimonious Learning-Augmented Caching studies how much prediction access is really needed [34, 35]. This perspective is complementary to LAFC-Evict rather than directly competing with it. Learning-augmented caching papers typically study online algorithms equipped with advice, predictor queries, and robustness objectives. By contrast, LAFC-Evict releases finite-horizon, continuation-policy-dependent counterfactual labels for every candidate within a decision. The artifact can train or diagnose predictors

and eviction-scoring models before they are embedded into an online algorithm, and it supports value regression, best-candidate selection, pairwise preference learning, and label characterization from one release.

10.5. Benchmark and Dataset Positioning

Public learning-to-rank collections such as LETOR and the Yahoo! Learning to Rank Challenge show how reusable labeled artifacts can organize a research community [36, 37, 38]. In that limited sense, LAFC-Evict is to learned cache eviction somewhat analogous to LETOR-style ranking corpora: a standardized candidate-level label surface reduces bespoke label generation and evaluation drift. The label design in LAFC-Evict also connects to counterfactual learning and ranking because each cache decision induces a candidate set, a minimum-loss subset, regret summaries, and a natural pairwise structure [39, 40, 41, 42]. Even so, LAFC-Evict is not simply a ranking dataset transplanted into systems. Its labels are tied to finite-horizon eviction outcomes under a specified continuation policy. The central differentiation to maintain throughout is therefore cautious but concrete: to our knowledge, we are not aware of a public cache-eviction artifact that releases reusable candidate-level finite-horizon counterfactual supervision together with decision-level and pairwise task views at this scale. In that sense, LAFC-Evict is neither a new online policy nor merely a raw trace repository or simulator-centric contribution, but a reusable benchmark artifact for learned cache-eviction research that complements existing traces, simulators, and policy papers.

11. Conclusion

This paper positions LAFC-Evict as a benchmark artifact for learned cache eviction rather than as a new policy paper. The core contribution is a release-oriented package built around candidate-level counterfactual supervision, derived decision and pairwise tasks, and validation-aware artifact construction. The preserved open release now couples manifest-backed scale and successful end-to-end validation with committed summaries of candidate-label distributions, a lightweight linear value-regression baseline, a derived best-candidate selector, and pairwise-sample-backed feature baselines that show measurable decision-relevant signal in the released candidate features.

Those results warrant conservative interpretation. The linear regression baseline is weak, the best-candidate selector is a reproducible benchmark floor

rather than deployment evidence, the pairwise results remain backed by the shipped capped sample rather than any full quadratic pairwise universe, and the labels remain finite-horizon continuation-policy-dependent counterfactuals rather than globally optimal online targets. Future extensions can focus on richer per-family and per-horizon characterization, stronger learned baselines, online policy studies that sit above the benchmark layer, and broader public hosting where redistribution constraints permit it.

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Declaration of competing interest

Data availability

The evaluated LAFC-Evict open-release snapshot (candidate rows, decision view, pairwise sample, manifest, and checksums; Section 6) is developed in the project’s public GitHub repository at <https://github.com/SoroushVahidi/lafc-evict-dataset> and archived under Zenodo concept DOI 10.5281/zenodo.21895843, released under a CC0-1.0 license, per the repository’s `CITATION.cff`.

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CRediT authorship contribution statement

Soroush Vahidi: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors used generative AI and AI-assisted coding and search tools to support manuscript drafting and editing, related-work exploration, code development, debugging, experiment organization, and result and table summarization. All scientific claims, related-work interpretations, code, experiments, tables, and final manuscript text were reviewed and verified by the authors, who remain fully responsible for the content and correctness of this work. No AI tool is listed as an author or as a contributor to authorship.

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